

Assessing Synthetic Benchmark Reliability: A Case Study on the Canon Layer

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Abstract

Synthetic reasoning tasks on graphs, such as multi-hop lookup and graph search, are an attractive way to test changes in transformer architecture: they are cheap to run and seem to isolate a clean notion of “reasoning ability”. We ask whether they can actually be trusted for this purpose, using the recently proposed Canon layer, which is a causal convolution inserted inside each transformer block, as a running example. We find that these benchmarks require careful use: tasks differ substantially in stability, some are reliable across multiple runs and others volatile enough that the random seed alone can flip which architecture looks better. Despite this variability, the benchmarks remain a useful tool for studying the mechanistic differences between architectures, since internal training signatures reproduce across seeds even where final scores do not. Accounting for these pitfalls, we evaluate CANON and find it improves reasoning performance consistently at every scale from 3M to 100M parameters. We then use the benchmarks to characterise CANON’s mechanistic signatures: it suppresses the deep-layer gradient dominance the baseline develops, is structurally biased toward integrating longer-range context from initialisation, amplifies residual-stream representations in the deep layers, and requires cross-token mixing to be effective. Removing it degrades performance below the baseline.

1 Introduction

Recent work in transformer architecture research has introduced a large number of new modifications and components: variations of attention mechanisms [DeepSeek-AI, 2024, Yang et al., 2023], positional encodings [Su et al., 2024, Kazemnejad et al., 2023, Press et al., 2022], and various gating mechanisms [Shazeer, 2020, Shazeer et al., 2017, Fedus et al., 2022]. There is no consensus on which components are worth keeping. The most significant challenge in architecture research is the scale and cost of training and evaluation. Synthetic benchmarks have emerged as an alternative to evaluate architecture changes in a cost-effective manner [Allen-Zhu and Li, 2023a, 2024, 2023b, 2025]. These benchmarks are small, fully controllable tasks such as arithmetic reasoning [Ye et al., 2024], synthetic grammar generation and various tasks on graphs.

These benchmarks are cheap to run, but there is no consensus on how to use them and no established trust in their verdicts. It is unclear whether a score difference between two architectures reflects a real architectural effect or an artifact of the particular task, seed, or scale. We therefore ask a concrete question: can these benchmarks be used for architecture search, and if so, how must they be run for their comparisons to hold up?

We develop both points using the CANON layer [Allen-Zhu, 2025] as a running example. Canon is a lightweight, depthwise one-dimensional causal convolution inserted at fixed points inside every

transformer block; it adds a second channel of token interaction alongside attention, mixing each position with a few of its immediate predecessors. The original work reports that Canon improves the reasoning depth of language models on these benchmarks, but leaves three questions open. It does not explain why the improvement emerges, nor under which conditions it holds rather than vanishing; and it does not examine how Canon’s performance and behaviour depend on its parametrization, such as the kernel size, the insertion positions, and the convolution’s initialization. We answer these questions, and in doing so demonstrate how the benchmarks should be used.

Contributions.

- We show the suite is not uniformly reliable. DEPO and BREVO grok at a seed-dependent step and cannot be trusted from a single run, while the graph-search tasks (BFS, Shortest-Path, ConComp) are stable across seeds. The seed-to-seed variance of the final score concentrates almost entirely on the grokking tasks.
- Averaged over seeds and swept across model sizes, CANON reaches lower loss than a strong LLAMA baseline at every scale from 3M to 100M parameters.
- We use the benchmarks to study mechanism, since the training behaviour is seed-stable even where the final score is not. We characterise CANON’s mechanistic signatures: suppression of deep-layer gradient dominance, amplification of residual-stream representations in deep layers, and a structural bias toward integrating longer-range context present from initialisation. Cross-token mixing is the essential ingredient: $k=1$ (no mixing) performs *worse* than the baseline, and the gain grows monotonically with kernel size.
- We conclude that architectures should be ranked only after averaging over seeds and scales, and that mechanism, which these benchmarks report reliably, is what they are best used for.

2 Background and Problem Statement

The Canon layer. A standard transformer block updates a residual stream purely along the feature dimension between attention operations; tokens interact only through attention. The Canon layer adds a second, lightweight channel of token interaction: a depthwise one-dimensional causal convolution with a small kernel of size k , applied along the sequence so that each position mixes in a few of its immediate predecessors. Formally, at each insertion point the hidden state h_t is updated as

$$h_t \leftarrow h_t + \sum_{j=0}^{k-1} w_j \odot h_{t-j}, \quad (1)$$

where $w_j \in \mathbb{R}^d$ are learned weights shared across the sequence and $\odot$ denotes elementwise multiplication. The explicit residual in (1) ensures the convolution can be a no-op and preserves the baseline as a special case. Causality ($j \geq 0$) preserves the autoregressive structure. A convolution can be inserted at up to four positions within a block (pre-attention, post-attention, pre-FFN, and post-FFN, labeled A through D; Figure 1), and we write CANON for the full A/B/C/D configuration unless noted. Our baseline, LLAMA, is the same architecture with the Canon convolutions removed.

Tasks and metrics. The benchmarks are graph-reasoning problems presented as token sequences. DEPO tests multi-hop knowledge retrieval (following a chain of k pointers), scored by hop- k accuracy. BFS tests breadth-first traversal (set recall), Shortest-Path tests path-finding (set accuracy), and ConComp tests connected-component factorization (prefix accuracy). BREVO tests reasoning breadth over a DAG and is excluded from the training mix because it is too unstable to train reliably. Each task is presented in two encodings, an edges-list and an adjacency-list form, which we treat as separate tasks. The standard training signal is the 8-task mix: the four core tasks in both encodings.

Configuration. Unless stated otherwise the default model has 30M parameters (8 layers, width 512), uses kernel size $k = 4$ at all four positions, and trains on the 8-task mix. For the scaling study we train at 3M, 10M, 30M, and 100M parameters with five seeds per (architecture, size) cell. The single metric we use for cross-architecture comparison is the held-out language-modelling loss (loss/out); per-task accuracies are used for the task-level breakdowns.

3 Experiments

3.1 Reliability of Synthetic Benchmarks

A single training run is not sufficient to make reliable architectural comparisons on these benchmarks. Text modelling is stable across seeds, but DEPO and BREVO exhibit grokking [Power et al., 2022]: the loss holds a plateau before dropping sharply, and the step at which this occurs is seed-dependent. The timing difference is large enough that a single CANON seed can finish above or below a single LLAMA seed depending entirely on initialization. The instability is localized: pooling five seeds per architecture at 30M, the seed-to-seed spread of the final score concentrates almost entirely on DEPO, while every graph-search task (BFS, Shortest-Path, ConComp) remains stable across seeds (Figure 2).

Any ranking claim on these benchmarks must therefore average over seeds, and since the gap between architectures can change with model size, over several scales.

The volatility of DEPO does not mean the depth-of-reasoning skill it probes is itself unmeasurable. Across the 40 scaling runs (both architectures $\times$ four scales $\times$ five seeds), of all the stable graph-search tasks ConComp is the one whose accuracy tracks DEPO grokking most strongly: whether a run has grokked DEPO is most predictive of its ConComp score. ConComp exercises the same global, multi-step composition over the graph but without the sharp grokking transition, so it serves as a more stable proxy for the same depth-of-reasoning ability.

3.2 Canon Consistently Improves Performance Across Seeds and Scales

Averaged over five seeds per point, CANON achieves lower loss than LLAMA at every scale from 3M to 100M parameters (Figure 3). The gap is largest for small models and narrows with scale, but the ordering never reverses. At 30M the gain is concentrated on DEPO, where CANON adds a large accuracy margin on both encodings; improvements on ConComp are minor, and BFS and Shortest-Path are already solved by both architectures. Notably, CANON’s benefit is largest on the same volatile, grokking-prone task identified in Section 3.1, which is why single-seed evaluations of it are so easy to misread: grokking, seed variance, and architectural benefit all coincide on multi-hop retrieval.

3.3 Canon Substitutes for Model Depth

The scaling sweep grows the model by widening it. A complementary question is how the two architectures respond to *depth*. We train a param-matched depth sweep at roughly 85M parameters, varying the number of layers while shrinking the width to hold the parameter count fixed (10 layers at width 832, 12 at 768, 14 at 704), with both architectures trained on the 8-task mix.

LLAMA improves steadily as it is made deeper: its loss falls from 0.181 at 10 layers to 0.103 at 14 (Figure 4, left). The improvement is concentrated on multi-hop retrieval: a 10-layer LLAMA is at the floor on DEPO (hop-4 accuracy 0.16) and only solves the task once it has 14 layers (0.93). Depth is what lets the baseline acquire the multi-hop capability.

CANON reaches the same capability several layers shallower. A 10-layer CANON model already attains DEPO accuracy 0.86 and a lower aggregate loss (0.141) than a 12-layer LLAMA, and a 12-layer CANON model (0.127) is the strongest configuration of either architecture, matching a 14-layer LLAMA on DEPO (Figure 4, right). The cross-token mixing Canon adds supplies what LLAMA otherwise obtains only by stacking more layers, which is consistent with the gain being largest on the task that most requires composing information across positions. The depth points use a single seed per configuration and DEPO is grokking-prone (Section 3.1), so we read the trend rather than the individual points; the direction is consistent across both depths at which CANON is available.

3.4 Canon Configuration: What Matters and What Does Not

The previous section establishes that CANON helps, but not which parts of its configuration are responsible. We examine two design choices that are easy to vary: where the convolution is inserted and how its weights are initialised.

Insertion positions: A and C are sufficient. A transformer block offers four insertion points (A through D, as described in Section 2). The full A/B/C/D configuration is not necessary: placing convolutions only at positions A (pre-attention) and C (pre-FFN) recovers essentially the entire gain (Figure 5, left). The B and D positions (post-attention and post-FFN) contribute negligibly and can be dropped without loss. This makes CANON a minimal, two-point intervention on the standard transformer block.

Initialisation: the benefit is robust. All initialisation schemes tested beat the LLAMA baseline, including zero initialisation (Figure 5, right). Zero-init starts as a no-op and recovers the benefit through learning; the $1/k$ constant-variance scheme (`const_var_sqrt`) is best but the gap over other schemes is small. The structural benefit of the convolution does not depend on the initialisation choice.

3.5 Mechanistic Signatures of Canon

The averaged ranking settles whether CANON helps but not why. Section 3.1 supplies the tool: although the final loss varies seed to seed on the grokking tasks, the changes an architecture makes during training (the distribution of gradients across depth, the scale of representations, the structure of the learned convolution weights) reproduce across seeds. A few runs already expose how two architectures differ mechanistically. The deep/shallow gradient ratio makes this concrete: where the final DEPO score swings seed to seed, the trajectory of this ratio is reproducible (Figure 6). The CANON seeds hold it low and tightly bundled throughout training, while the LLAMA seeds both drift upward, and the two bundles never cross. Even where the benchmark cannot rank the two

models, it offers a seed-stable window into how they differ internally. We examine these signatures and then probe which structural property of the convolution is essential to the gain.

3.5.1 Across-Layer Training Dynamics

LLAMA develops a strong deep-layer gradient dominance over training, with deep layers increasingly dominating the gradient signal; CANON suppresses this effect (Figure 7). CANON also amplifies residual-stream representations most strongly in the deep layers near the language-modelling head, and distributes the per-layer gradient profile more evenly across depth. These signatures are consistent across seeds, making them reliable descriptors of how CANON reshapes the model’s internal computations.

3.5.2 Cross-Token Mixing Is the Essential Ingredient

Setting $k=1$ removes cross-token mixing while preserving all other CANON parameters; this configuration performs *worse* than the LLAMA baseline, and loss improves monotonically as k increases from 2 to 6 (Figure 8). Cross-token mixing is the necessary ingredient, not the additional parameters. The learned weights place more mass on farther-back shifts than on the current token from the first checkpoint, and the mixing is strongest in the early layers. CANON is therefore structurally biased toward integrating context from earlier tokens, a property present from initialisation rather than acquired gradually.

4 Discussion

On Canon. CANON consistently outperforms the LLAMA baseline at every scale tested, with the gain concentrated on multi-hop retrieval (DEPO). The causal convolution provides cross-token mixing that is structurally biased toward long-range context from initialisation: removing it ($k=1$) degrades performance below the baseline, confirming mixing is the essential structural ingredient. CANON also produces reproducible across-layer signatures (gradient redistribution and deep-layer representation amplification) that are consistent across seeds and reflect how the convolution reshapes the model’s internal computation.

Methodological lesson. These benchmarks mislead when read one seed at a time, because grokking makes a single training curve on the volatile tasks largely a function of the seed. The fix for ranking claims is to average over seeds and sweep scale; it is simple but not optional, and which tasks demand it is not obvious a priori. The benchmarks also report two things with very different reliability: the final score on a grokking task is a weak signal, while the training dynamics reproduce even where the score does not. The benchmarks are therefore better used to study mechanism, with the seed-averaged score serving as a coarse sanity check rather than the object of study.

Limitations and future work. Our mechanistic evidence is observational: the signatures we document are descriptive and do not establish causal direction. Direct interventions (frozen or surgically modified convolution weights, or targeted gradient-flow modifications) remain future work and would help pin down the mechanisms more precisely. We also leave the dependence on encoding (edges-list versus adjacency-list) unexplained. Finally, the suite we study is a specific one; whether the reliable/unreliable split we document generalises to other synthetic benchmark families is an open question.

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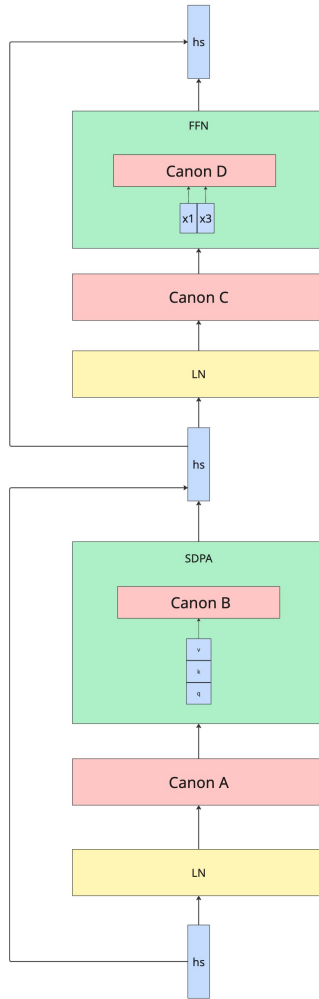


Figure 1: **Canon insertion points.** A depthwise causal convolution can be placed at four positions within a transformer block: pre-attention (A), post-attention (B), pre-FFN (C), and post-FFN (D). CANON denotes the full A/B/C/D configuration; the baseline LLAMA removes all four.

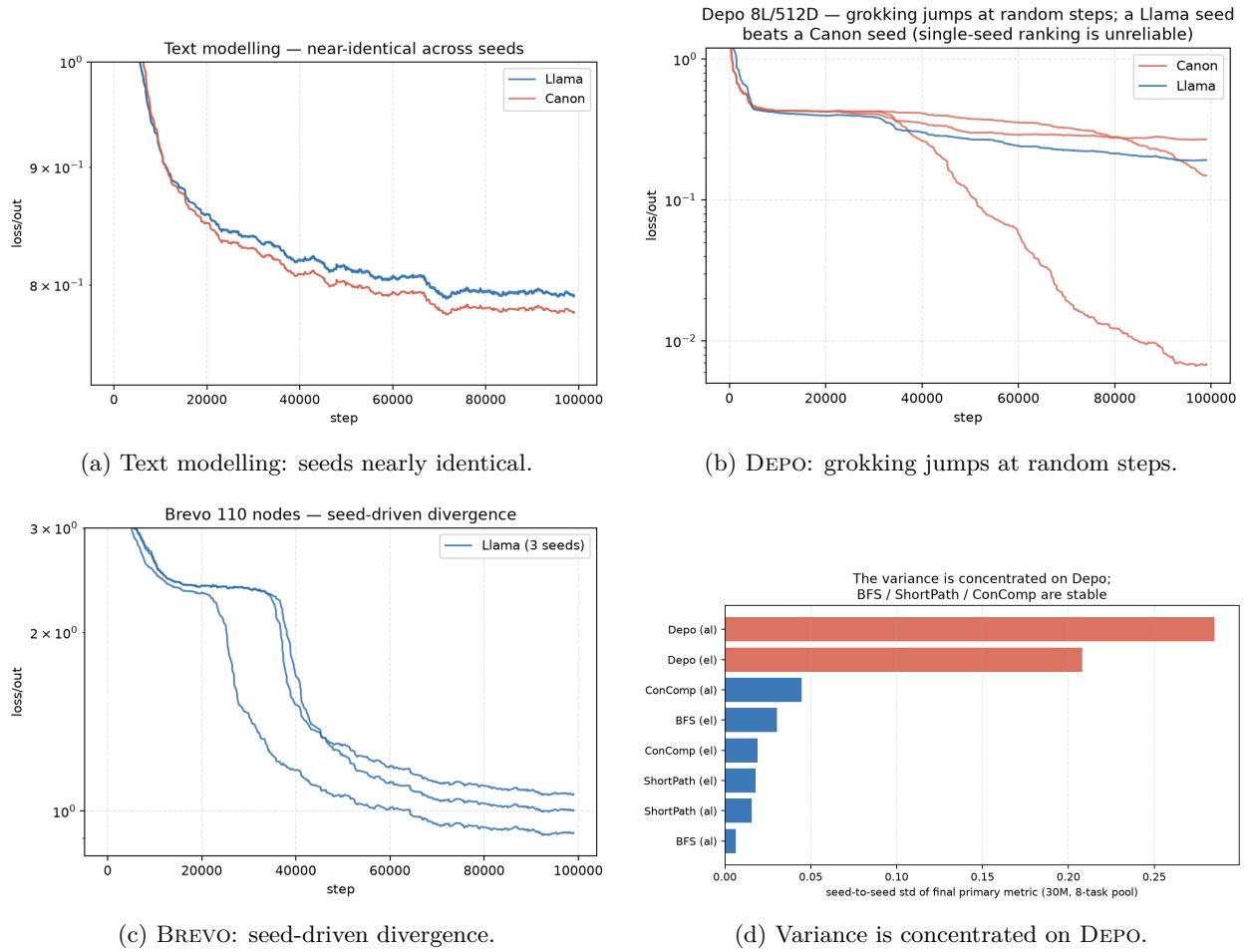


Figure 2: **Benchmark reliability varies by task.** Text modelling is stable across seeds; DEPO and BREVO grok at a seed-dependent step, making single-run rankings unreliable. Seed-to-seed variance concentrates on DEPO; all graph-search tasks are stable.

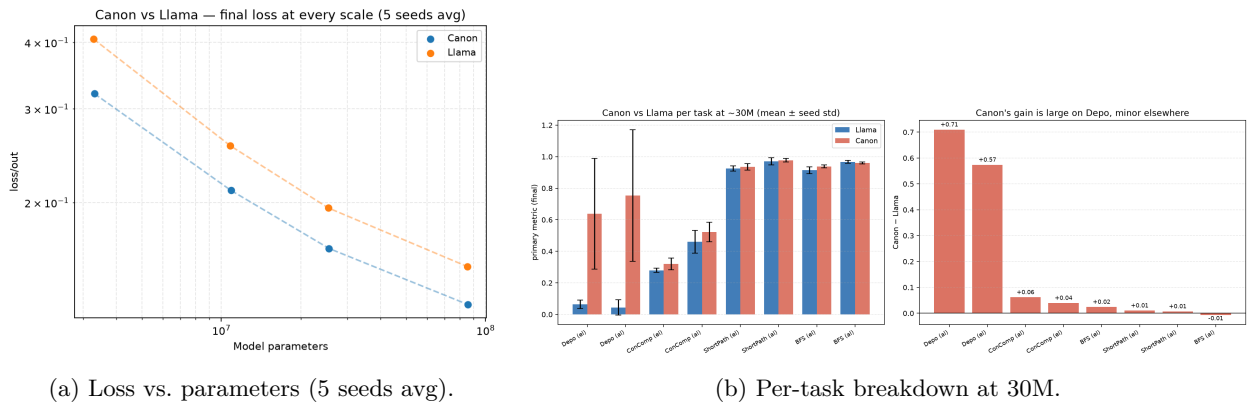


Figure 3: **Canon outperforms Llama at every scale when averaging over seeds.** The gap is largest for small models and narrows with scale. At 30M the gain is concentrated on DEPO; improvements on other tasks are minor or absent.

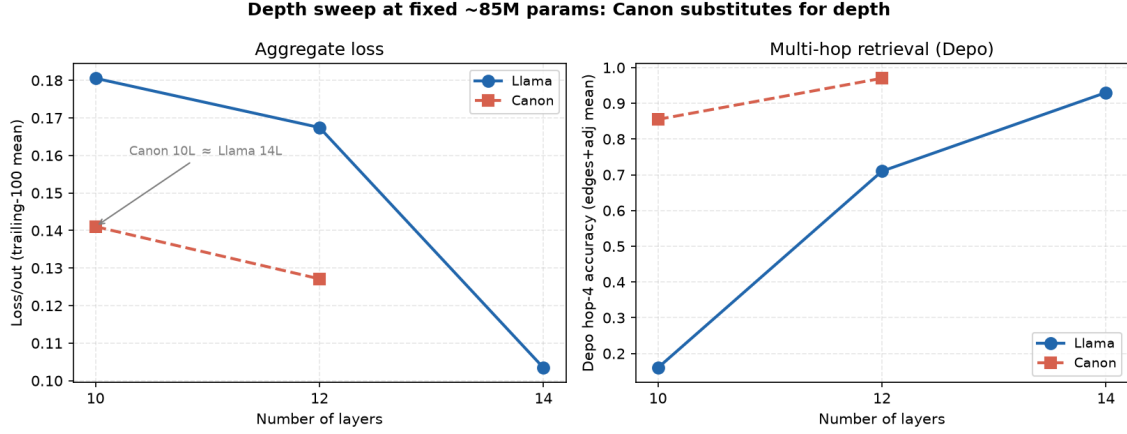
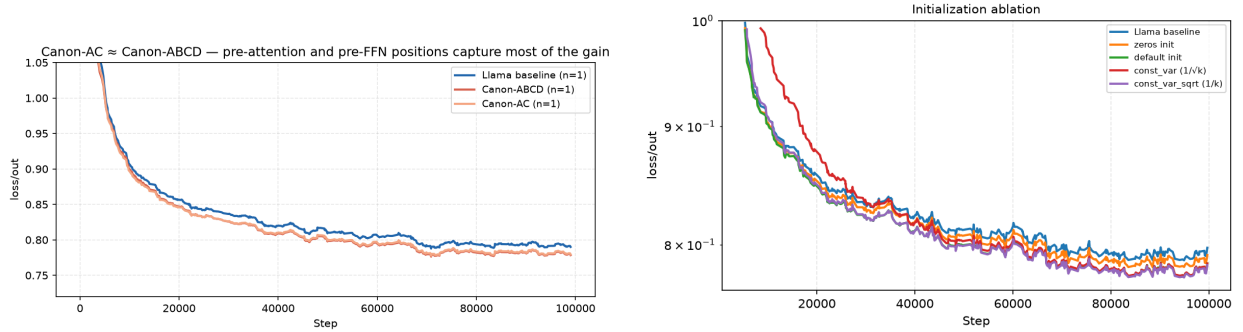


Figure 4: **Canon substitutes for model depth (param-matched, ~85M)**. LLAMA improves with depth, acquiring multi-hop retrieval (DEPO) only at 14 layers; CANON reaches comparable performance several layers shallower. CANON at 10 layers matches a 14-layer LLAMA on DEPO, and CANON at 12 layers is the strongest model overall.



(a) Canon-AC matches Canon-ABCD.

(b) All init schemes beat the baseline.

Figure 5: **Canon configuration ablations**. Positions A and C (pre-attention and pre-FFN) capture the full gain; B and D are droppable. All initialisation schemes outperform the LLAMA baseline, including zero initialisation; the benefit is structurally inherent to the convolution, not to the choice of starting weights.

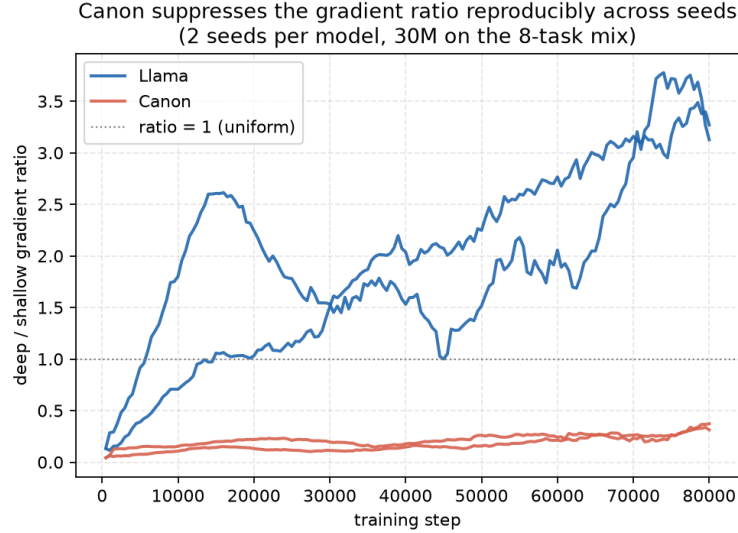


Figure 6: **Mechanism is reproducible where the score is not.** Deep/shallow gradient ratio over training, one line per seed. The CANON seeds (held low and tightly bundled) and the LLAMA seeds (drifting upward) form two non-overlapping bundles, even though the final DEPO score varies seed to seed within each architecture.

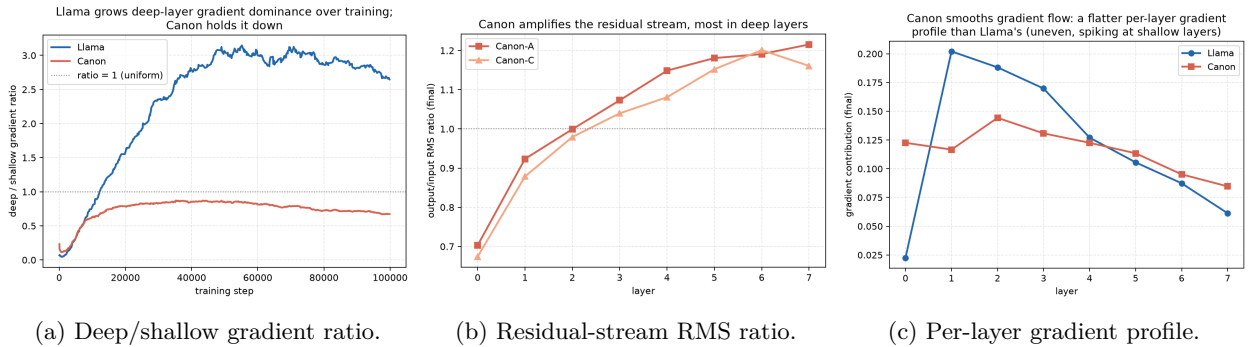


Figure 7: **Canon's across-layer reorganization signature.** CANON suppresses the deep-layer gradient dominance that LLAMA develops, amplifies deep-layer representations, and distributes the per-layer gradient profile more evenly, changes that resemble the effect of scaling the model up.

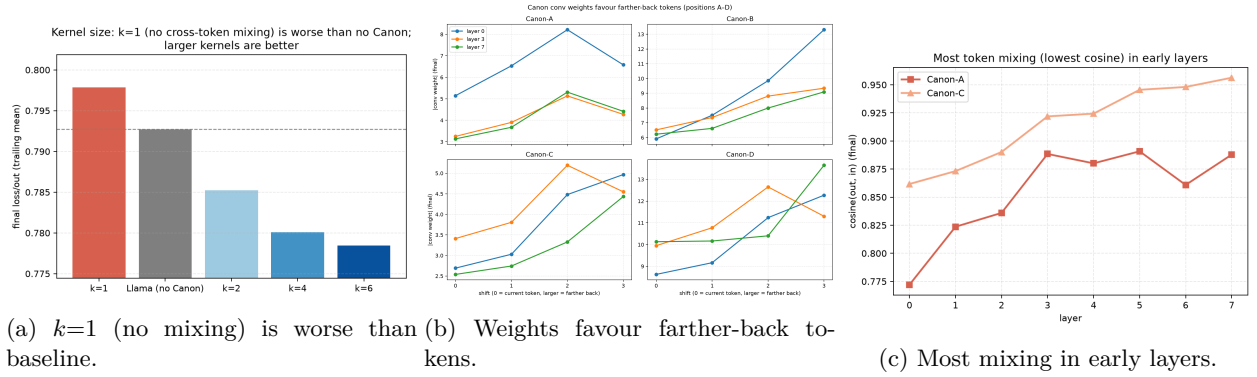


Figure 8: **Cross-token mixing is the necessary ingredient.** $k=1$ (no mixing) performs worse than the baseline; loss improves monotonically with kernel size. The learned weights favour farther-back tokens from initialisation, with the strongest mixing in early layers.